مدارس الإمارات الوطنية
Emirates National Schools

Grade: 4 | Term 1, Week 11 (9 Nov to 13 Nov) | TCD: Future Skills 4: Design, Tools and Simple Solutions | Session length: 1 hr 30 min

T1 WEEK: 11	**Lesson 1: Function Blocks**
Standard(s):	Technology: decomposition and abstraction (grouping repeated steps); Math: 4.OA.C.5 (generate and analyse patterns)
Literacy Strategies:	Pattern hunt with coloured cards; sentence frame: My function holds the steps ... and I call it ... times.
Assessment:	Before and after code comparison on the worksheet; observation; live quiz score on the TCD website.
WALT (Learning Target):	We are learning to use a function block to make our code shorter and easier to read.
WILF (Success Criteria):	I can find a pattern of steps that repeats in my code. I can put the repeated steps into a function. I can call the function to run those steps. I can explain why the shorter code is better.
Targeted Vocabulary	function, pattern, repeat, call, sequence, efficient, debug
Resources	Matatalab Pro Sets, Function blocks and function card, Map mats, Artist add-on and pens, Coloured pattern cards, A3 paper Accompanying resource: Grade4_Week11_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade4_Week11_Worksheet and Grade4_Week11_Lesson_Plan.
National Identity Integration	Repeating patterns in Sadu weaving: Emirati weavers repeat the same small pattern to build a long, beautiful design, just like a function repeats steps.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Explain what a function is and how to call it (slide 6). Model finding the pattern in a long code using coloured cards (slide 7). Build the square with a function and compare block counts (slides 8 and 9).
Guided Practice: We Do	Class looks at a long staircase code on the board and agrees which steps repeat. Pairs build the staircase function together with teacher support.
Independent Practice: You Do	Pairs complete the function challenges and record long and short code (slide 11). Challenge levels on the board (slide 12). Live quiz on tablets (slide 13).
Differentiated Instruction: Beginning / On / Above	**Beginning: Use the function card to draw the square, copying the example.** **On: Create functions for the square and the staircase yourself.** **Above: Use a function together with a loop and draw a pattern of two squares.**
Closure/Reflection	Pairs share their block count before and after. Exit ticket: why is a function useful? (slide 14)
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☐ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☐ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☑ Scaffold / Sentence Starters

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G&T / Extension	☑ Extension / Challenge Task ☐ Real-World Application ☑ Higher-Order Questioning ☐ Curriculum Compacting ☐ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____